Quantitative Methods I

Week 6 - Functions with Multiple Variables

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Motivation

- Real-world business decisions are rarely about a single variable.
- Production depends on
  - capital and
  - labor
- Consumer satisfaction (utility) depends on
  - a bundle of different goods
- But, how do we model and analyze such multi-variable relationships?

Motivation - cont'd

A company produces goods using both capital (K, e.g., machinery) and labor (L, e.g., employee hours). Let the **Cobb-Douglas function** model this production:

$$Q(K, L) = 5\sqrt{K \cdot L}$$

Key Business Questions:

- If we have 4 units of capital and 9 units of labor, what is our output?
- How can we visualize the relationship between K, L, and Q?
- If we can only afford to increase *one* input, which one gives us a bigger "bang for our buck"? Labor or Capital?

To answer these, we need to understand **functions with multiple variables**.

Motivation - cont'd

Today, we build a toolkit to model and analyze relationships with multiple inputs.

The Question	**The Tool**
• How do we model outputs that depend on multiple inputs?	→ Functions of two (or more) Variables
• How can we visualize a 3D relationship on a 2D map?	→ Level Curves (Indifference/Isoquant Curves)
• How do we measure the impact of changing just *one* input?	→ Partial Derivatives (Marginal Analysis)

Functions of Two Variables

- **A function of two variables** x and y with domain D is an arithmetic rule that assigns a unique value $z = f(x, y)$ to each pair (point) (x, y) in D.
- x and y are the **independent variables**; z is the **dependent variable**.
- The range R is the set of values z can attain

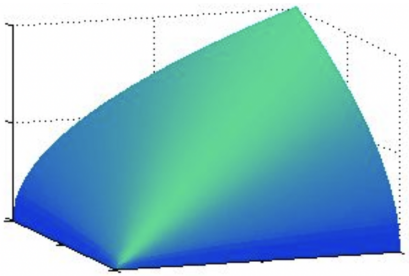
Our Production Model:

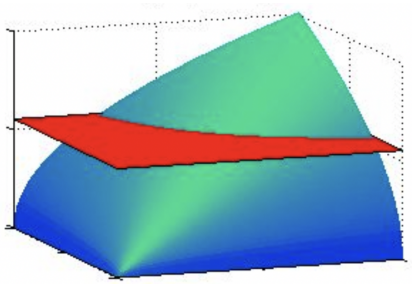
- $Q(K, L) = 5\sqrt{K \cdot L}$ is a function of two variables
- The inputs (independent variables) are Capital (K) and Labor (L)
- The output (dependent variable) is Production Quantity (Q)
- For K=4 and L=9, the output is

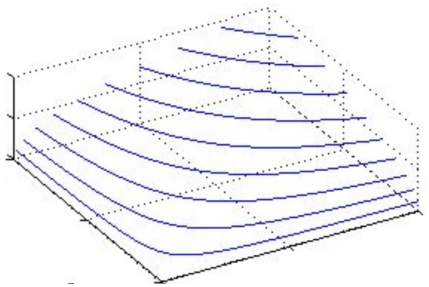
$$Q(4, 9) = 5\sqrt{4 \cdot 9} = 5\sqrt{36} = 30$$

Visualizing Functions of Two Variables

A function of two variables creates an impressive, but hard-to-work-with 3D surface







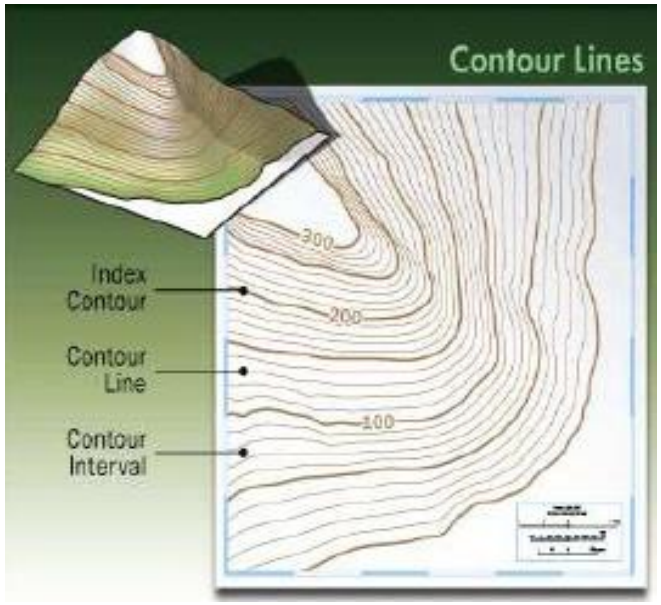
A more practical approach is to view the surface from above using **Level Curves**, which connect all input pairs (K, L) that produce the *same constant output level*.

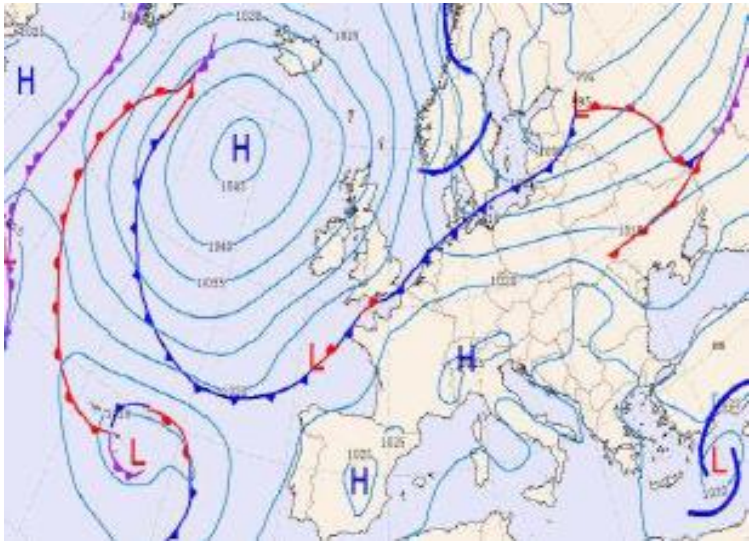
- For production functions, these are called **isoquants** ("same quantity")
- For consumer utility functions, these are called **indifference curves**

Level Curves

LEVEL CURVE

For a function f and a given number c, the curve consisting of all domain points (x, y) for which it holds that $f(x, y) = c$ is called the level curve of f at level c.



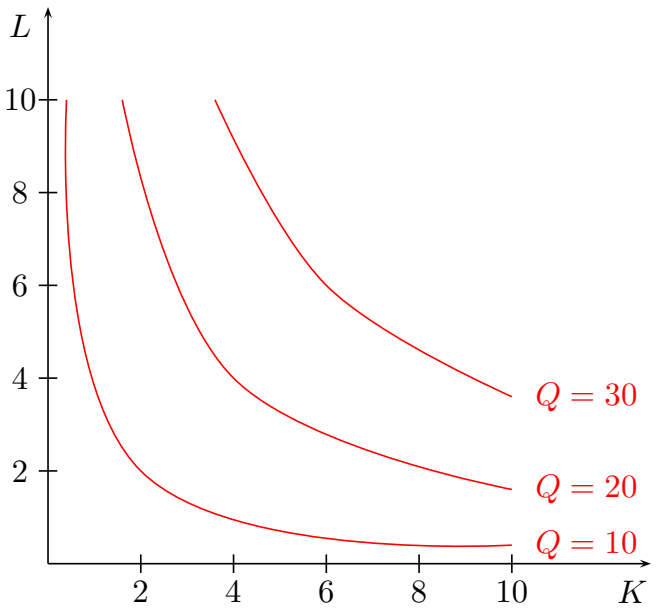


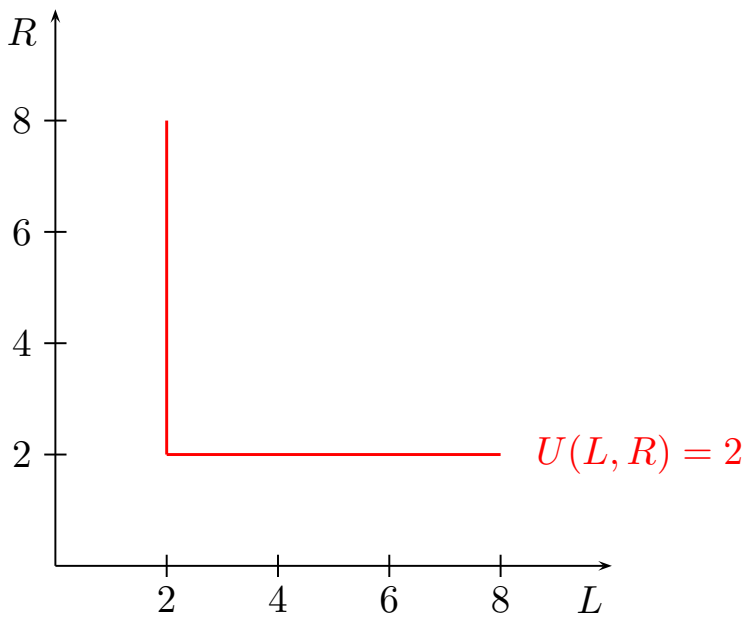
Contour Lines of Mountains (geography class) and Isobars (weather forecasts)

Isoquants and Indifference Curves

Isoquants at level c, for our production function $Q(K,L) = 5\sqrt{KL}$,

$$Q(K,L) = c \;\Leftrightarrow\; 5\sqrt{KL} = c \;\Leftrightarrow\; \sqrt{KL} = \frac{c}{5} \;\Leftrightarrow\; KL = \frac{c^2}{25} \;\Leftrightarrow\; L = \frac{c^2}{25}K^{-1}$$





Indifference curves at level c, for utility function $U(L,R) = \min\{L,R\}$,

$$U(L,R) = c \;\Leftrightarrow\; \min\{L,R\} = c \;\Leftrightarrow\; L = c \text{ or } R = c$$

Partial Derivatives

PARTIAL DERIVATIVE

Let f be a function of x and y.

The derivative of $f(x, y)$ with respect to x, treating y as a constant, is called the partial derivative of f with respect to x. It is denoted by $f'_x(x, y)$.

The derivative of $f(x, y)$ with respect to y, treating x as a constant, is called the partial derivative of f with respect to y. It is denoted by $f'_y(x, y)$.

Application: Marginal Productivity

Let's apply this to our production function: $Q(K, L) = 5\sqrt{KL} = 5K^{1/2}L^{1/2}$

- **Partial derivative w.r.t. Capital (K):** Treat L as a constant.

$$Q_K = \frac{\partial Q}{\partial K} = 5 \cdot \left(\frac{1}{2}K^{-1/2}\right) \cdot L^{1/2} = 2.5\frac{\sqrt{L}}{\sqrt{K}}$$

  This is the **Marginal Product of Capital**, i.e., the extra output from one more unit of capital, holding labor constant.

- **Partial derivative w.r.t. Labor (L):** Treat K as a constant.

$$Q_L = \frac{\partial Q}{\partial L} = 5K^{1/2} \cdot \left(\frac{1}{2}L^{-1/2}\right) = 2.5\frac{\sqrt{K}}{\sqrt{L}}$$

  This is the **Marginal Product of Labor**, i.e., the extra output from one more unit of labor, holding capital constant.

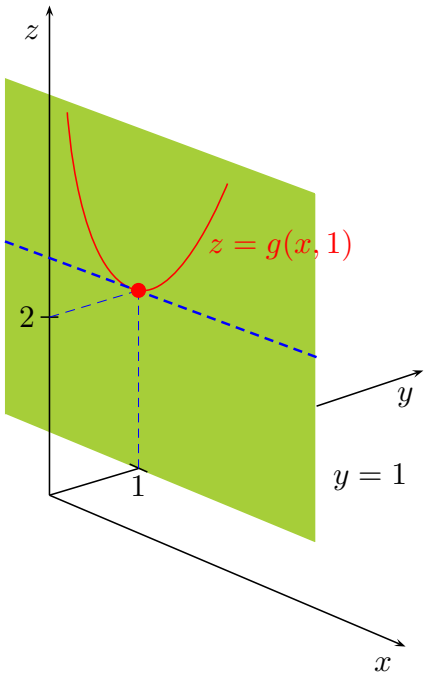
Picturing Partial Derivatives

Consider the function $g(x, y) = x^2y + y^2 + y$.

The partial derivative of g with respect to x is

$$g'_x(x, y) = 2xy$$

Now, let us zoom in on $g'_x(0, 1)$. Note that $g'_x(0, 1)$ is the derivative of the function $g(x, 1) = x^2 + 2$, which is a vertical 'slice' of the graph of the original function g, for $y = 1$. (Green square: all points with $y = 1$)



Approximations

	a linear function ℓ	*a nonlinear function* f
$x : a \to a + \Delta x$ $y : y = b$	$\Delta z = \ell'_x(a,b)\,\Delta x$	$\Delta z \approx f'_x(a,b)\,\Delta x$
$x : x = a$ $y : b \to b + \Delta y$	$\Delta z = \ell'_y(a,b)\,\Delta y$	$\Delta z \approx f'_y(a,b)\,\Delta y$

Elasticity

ELASTICITY

If f is a positive function of x and y, then the elasticity $\mathrm{El}_x(x,y)$ of f ($x > 0$ and $y > 0$) with respect to x is defined as

$$\mathrm{El}_x(x,y) = f'_x(x,y) \cdot \frac{x}{f(x,y)}$$

and the elasticity $\mathrm{El}_y(x,y)$ of f with respect to x as

$$\mathrm{El}_y(x,y) = f'_y(x,y) \cdot \frac{y}{f(x,y)}$$

Our Production Model:

$$Q(K,L) = 5\sqrt{KL}$$

For the elasticity of this production function Q w.r.t. capital, we have

$$\mathrm{El}_K(K,L) = Q'_K(K,L) \cdot \frac{K}{Q(K,L)} = 2.5\frac{\sqrt{K}}{\sqrt{L}} \cdot \frac{K}{5\sqrt{KL}} = .5$$

Higher order partial derivatives

Let

$$f(x,y) = x^2y + 3x - y^2$$

Then, the first order derivatives are

$$f'_x(x,y) = \frac{\partial f}{\partial x} = 2xy + 3 \qquad \text{and} \qquad f'_y(x,y) = \frac{\partial f}{\partial y} = x^2 - 2y$$

As such, the second-order derivatives are

$$f'_{xx}(x,y) = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = 2y \qquad f'_{yx}(x,y) = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = 2x$$

$$f'_{xy}(x,y) = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = 2x \qquad f'_{yy}(x,y) = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = -2$$